GOOGLE GEMMA 4 DEVELOPER AGENT COMPETITION · PAPER TRACK (TASKS, BENCHMARKS & NEW RESOURCE)

qb-cubie in a Test-Guided Coding Agent: Methodology, Development, and Exploratory Pilot Findings

A Zero-Leakage A/B Evaluation of Example-Based Testing (qb-cubie 0.0.3) vs. pytest on Everyday Consumer Hardware Across Single-Agent (v1–v5) and Multi-Agent (v6–v9) Architectures

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Data Cutoff: November 12, 2026 (v1–v5: Nov 11; v6–v9: Nov 12) · Artifacts: Prepared for public release on PyPI (qb-cubie 0.0.3) and GitHub (MIT License, commit 5e50d22f80db90c4e8ae6e3aeedab6acdb981230).

ABSTRACT — Evaluating open-weights coding agents on everyday consumer hardware requires harnesses that separate genuine code repair from test manipulation. We present qb-cubie (v0.0.3), an open-source Python quality assurance framework providing declarative Example-Based Testing (EBT), and evaluate whether providing qb-cubie and its structured guide improves the resolution rate of a local gemma4:12b agent (11.9B, GGUF Q4_K_M via Ollama 0.40.2) on real-world SWE-bench Python tasks compared to standard pytest. Both arms run under identical budgets and repository snapshots, while an independent, Cubie-free container strips all agent test edits and evaluates only the production patch against private FAIL_TO_PASS and PASS_TO_PASS suites.

Across the single-agent series (v1–v5; 7 completed pilots, 42 runs on 3 tasks with Seed 42, backed by 5,563 verified SHA-256 hashes), private resolutions occurred in 3 of 42 runs (7.1%): twice with pytest (v2, v4) and once with qb-cubie (revised v3). However, EBT contracts were executed in only 2 of 21 Cubie runs (9.5%) and remained unused in 19 of 21 runs (90.5%). In Pilot v5 (think=true), both arms resolved 0/3 tasks: 11 of 67 model calls hit the 8,192-token cap emitting only thinking tokens without a JSON action, consuming 87.3% (90,112 of 103,235) of output tokens and 79.6% of agent runtime. A follow-up series across four serial multi-agent generations (v6–v9: Analysis, QA, Dev; 17 runs, 1,441 model calls, 8.55M input tokens) eliminated infrastructural errors in v9 (0 cursor errors, 0 tool errors, 0 context overflows), but removing v8's forced handoff while overloading the finish tool schema caused the model to invoke finish zero times (0 calls) across 1,184 steps (974 duplicates, 82.6%), locking execution in Analysis before the QA-scoped qb-cubie guide could be injected.

1. Introduction

Deploying autonomous coding agents on everyday consumer hardware using compact open-weights models such as gemma4:12b (11.9B parameters, 4-bit quantized Q4_K_M) exposes operational bottlenecks that are often masked by frontier cloud APIs or permissive benchmarks. When an agent must reproduce a bug, author a test, and patch production code, failure rarely stems from a single root cause; instead, inference limits, pre-training biases, noisy repository baselines, and tool-interface friction interact as a compound bottleneck cascade [8].

To investigate how test methodology interacts with these constraints, we introduce qb-cubie (v0.0.3, commit 5e50d22f80db90c4e8ae6e3aeedab6acdb981230) and conduct a cryptographically frozen A/B study asking [4]: Does access to qb-cubie and its specified guidance improve the independently verified resolution rate of a test-guided coding agent powered by Ollama and gemma4:12b under a fixed working budget compared to the exact same agent using standard pytest? Because the intervention pairs the library with its instructional prompt [6], the study evaluates the combined package offering. Furthermore, because upstream inference and orchestration bottlenecks prevented EBT execution in 90.5% of v1–v5 runs and 100% of v9 runs, this study serves primarily as a forensic architectural bottleneck diagnosis rather than a conclusive head-to-head test of EBT efficacy [4, 10].

1.1 Declarative Contract Semantics of qb-cubie and Epistemological Boundaries

The open-source package qb-cubie unifies declarative Example-Based Testing (EBT), natural-language Review-Based Testing (RBT), execution arenas (TempFS, PyVenv), static analysis orchestrators (AstLinter, CstLinter), and a benchmark subsystem (qb_cubie.benchmarks / cubie-bench) with native SWE-bench support [9]. An EBT contract defines a finite set of concrete executions of a callable target via mutable WorldState dataclasses (args, kwargs, arena, fs, returns, expect, raises, exception). Given input states P in pre, permitted patterns Q+ (post.expect=True), and forbidden patterns Q− (post.expect=False), where match(p, q) checks return values, expected exceptions via ExceptionExpectation, and filesystem diffs in fs, the finite evaluation rule is [9]:

```
step_ok(p) = (Q+ is empty OR at least one q in Q+ matches) AND (no q in Q− matches)
run_ok     = all step_ok(p) hold;   evaluation = run_ok if EBT.expect=True else NOT
run_ok
```

While the structure “pre-state – operation – post-state” is inspired by a Hoare triple {P} S {Q} [1], WorldState is a concrete data pattern rather than a formal state predicate [9]. Passing an EBT contract demonstrates observed conformance over finite inputs rather than a formal correctness proof. Because all inputs in pre are checked against the shared set post, correlated input/output pairs require separate EBT contracts (e.g., via pytest.mark.parametrize), and WorldState(returns=None) explicitly requires None whereas qb_cubie.UNSPECIFIED disables return checking [9]. In our study, an EBT contract counts as utilized only if executed via .run() and verified via assert result.evaluation is True [4].

An internal audit of the repository (2,569 passing tests in 550.50s, Python 3.13.12) disclosed three core edge cases that did not affect the agent runs and are scheduled for patching in version 0.0.4 [9]: (B1, P1) PyVenv stores IPC files (ipc_in.pkl, ipc_out.pkl, _cubie_phantom.py) inside the workspace; (B2, P1) PytestReporter relies on assert statements ignored under python -O; and (B3, P2) CstLinter.check() returns [] on missing paths. Following an early audit warning against circular validation, all 698+ verification nodes for the study controllers were written with standard pytest assertions rather than EBT contracts [9, 10].

2. Related Works

Repository-level coding benchmarks are anchored by SWE-bench, SWE-bench Lite, and SWE-bench Verified [2], which evaluate whether generated patches resolve real GitHub issues against Fail-to-Pass and Pass-to-Pass test suites. Agentic frameworks such as SWE-agent, AutoCodeRover, and Agentless [3] show that structured navigation, search, and localized editing interfaces strongly influence resolution rates, though most published baselines rely on frontier models. Our work extends this literature in three ways: (1) we introduce qb-cubie as a declarative EBT alternative to procedural pytest assertions [9]; (2) we enforce a zero-leakage evaluation boundary that permits agents to freely edit visible tests during test-guided repair while stripping all test edits prior to private container evaluation [4]; and (3) we provide a forensic failure taxonomy of a 4-bit quantized 12B model across nine single- and multi-agent architectures [8, 10].

3. Methodology, SWE-bench Dataset, and Zero-Leakage Harness

The study task pool comprises 129 real-world GitHub pull requests and issues formatted in the SWE-bench schema [2] (instance_id, repo, base_commit, patch, test_patch, problem_statement, hints_text) across four Python repositories: 67 from FastAPI (fastapi/fastapi), 48 from Rich (Textualize/rich), 13 from Requests (psf/requests), and 1 from HTTPX (encode/httpx) [5, 10]. In addition, cubie provides a native SWE-bench module under src/qb_cubie/benchmarks/swebench/ supporting official Hugging Face datasets [9]. A manual audit of 39 tasks [5] identified 26 qualified candidates, 9 conditional cases, 3 deferred cases, and 1 excluded case (90 unaudited). Three tasks were frozen with Seed 42 for the iterative pilot series [4]:

Task ID	Repository & Category	Required Modification	FAIL_TO_PASS	PASS_TO_PASS
fastapi_9425	FastAPI · API/Schema	Treat stringified return annotation "None" in get_typed_return_annotation as None so HTTP 204 bodiless responses succeed.	1	8
requests_6589	Requests · Encoding	Use UTF-8 encoded byte length instead of character length for strings in super_len (Content-Length calculation).	1	11
rich_3278	Rich · ANSI Parsing	Strip private ANSI escape sequences from ESC 0 to ESC ? so console output is not corrupted.	16	7

Before each run, snapshots are sanitized of git history and reference patches. Both arms (cubie and pytest) run serially from identical containers with alternating starting arms. Agents may freely edit production files and visible tests; afterward, a fresh, Cubie-free evaluator container strips all test and runner edits, applies only the production patch, and executes the frozen private FAIL_TO_PASS and PASS_TO_PASS suites [4]. A task is resolved if and only if all private tests pass.

Repository Autonomy and Cryptographic Integrity: The cubie package (pyproject.toml) has no dependency on external harnesses such as swegemma [10]. In early single-agent runs (v1–v5), a local shell wrapper (./study) invoked a neighboring virtual environment (../devagent-comp/.venv/bin/python) containing swegemma 0.2.11 for container bridging; from v6 onward (team_workspace.py, team_v9.py), workspace management runs autonomously via native git subprocesses and standard Docker containers [10]. Across all eight v1–v5 archives under runs/, 5,563 SHA-256 hash references across input manifests, preflight files, and run manifests were verified with 100% agreement [4]. In Pilot v5 (20261111-200641), gemma4:12b (digest 4eb23ef187e2c5462566d6a1d3bbbc2f1346d0b4327cbb66d58fffbcc9b2b05c) ran on Ollama 0.40.2 (Python 3.13.18, pytest 9.2.0, qb-cubie 0.0.3) with Seed 42, temperature 0.2, think=true, num_ctx=32768, num_predict=8192, 900 s agent limit (90 s command / 120 s verification), and 80,000 configured context bytes (40,090 effective bytes) [4].

4. Single-Agent Experiments (Generations v1–v5)

The single-agent core evolved across five frozen designs [6]: early unversioned (basic tools, 600 s / 24 calls), v2 (working ledger, anchored edits, output repair), v3 (action-coupled notes, baseline counter-checks, EBT tracking; expanded in revised v3 to 900 s, unlimited calls, and 32k/8k tokens), v4 (runner-bound file hashes, oracle checks), and v5 (mandatory analysis checkpoints plus think=true). Table 2 summarizes all seven completed pilots (42 runs) and one prepared-only archive (180104; a separate 60 s smoke check in local/live-pilot-validation is excluded) [4]:

Archive ID	Core & Thinking	Budget (Time / Calls · Ctx / Out)	pytest Resolved	Cubie Resolved
132401 & 160158	early · think=false (2 pilots)	600 s / 24 · 16,384 / 4,096	0/3 & 0/3	0/3 & 0/3
163110	v2 · think=false	600 s / 24 · 16,384 / 4,096	1 / 3	0 / 3
171515 & 180104	v3 · think=false (180104 prep. only)	600 s / 24 · 16,384 / 4,096	0/3 & —	0/3 & —
180702	v3 rev. · think=false (1 unavail.)	900 s / unlim. · 32,768 / 8,192	0 / 3	1 / 3
192412	v4 · think=false	900 s / unlim. · 32,768 / 8,192	1 / 3	0 / 3
200641	v5 · think=true	900 s / unlim. · 32,768 / 8,192	0 / 3	0 / 3

- **Verified Resolutions & Exit-Code-4 Sensitivity:** Only 3 of 42 runs (7.1%) achieved private resolution: pytest on requests_6589 in v2 (163110); Cubie on fastapi_9425 in revised v3 (180702, run 0001, executing 6 EBT contracts with 5 passing); and pytest on fastapi_9425 in v4 (192412, where the production patch passed all 9 private tests [−33.3 pp, CI [−100; 0]], though the agent overwrote a 1,114-line visible test file with 45 broken lines and stalled) []. In 180702 (run 0004, Requests), the Cubie agent ran 3 EBT contracts but introduced an indentation error into production code, causing pytest exit code 4 during private collection. Treating run 0004 as unavailable yields +50.0 pp (CI [0; +100]) on 2 complete pairs; counting this agent-caused syntax error as a failure yields +33.3 pp across 3 complete pairs [].
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- **90.5% EBT Non-Utilization & Baseline Pollution:** Across all 21 Cubie runs in v1–v5, EBT contracts were executed in only 2 runs (9.5%, both in 180702) and never executed in 19 runs (90.5%) [10]. Visible suites also contained pre-existing failures before agent edits [7]: FastAPI had 8 collection errors (missing pydantic_settings); Requests had 577 passed, 5 failed, 15 skipped; and Rich had 746 passed, 65 failed, 24 skipped.

4.1 Deep Dive into Pilot v5 (Archive 20261111-200641)

Run	Task & Arm	Agent Time	Model / Tool Calls	Termination & Final Patch
0001 / 0002	FastAPI · Cubie / pytest	727.4 s / 681.4 s	8/6 & 6/4	Both invalid_output · both empty patches
0003 / 0004	Requests · pytest / Cubie	829.3 s / 900.1 s	17/15 & 19/17	invalid_output & agent_timeout · test edits only
0005 / 0006	Rich · Cubie / pytest	653.5 s / 684.7 s	8/6 & 9/7	Both invalid_output · both empty patches

In Pilot v5 (0/3 vs. 0/3; degenerate CI [0; 0]; 4,476.4 s total agent time [74.6 min], 80.3 min study wall time; median 684.7 s pytest vs. 727.4 s Cubie; 67 model calls [32 pytest, 35 Cubie], 55 tool calls [26 pytest, 29 Cubie], 1 vs. 4 tool errors, 0 EBT executions), documented tokens totaled 506,573 (252,363 in pytest [198,821 in / 53,542 out] plus 254,210 partial in Cubie due to the run 0004 timeout; 403,338 input and 103,235 output overall) [4]. Crucially, 11 of 67 model calls hit done_reason=length at 8,192 tokens with empty message.content and only thinking output, consuming 90,112 output tokens—87.3% of all known output tokens and 79.6% of agent runtime [8]. On requests_6589, pytest (run 0003) wrote valid UTF-8 tests (super_len("€") == 3, super_len("Test€") == 7) that failed with 1 and 5, but its production edit failed with edit_conflict/anchor_missing; Cubie (run 0004) corrupted 🚀 into two UTF-16 surrogates ('\ud83d\ude80', length 2) in a plain assert without EBT, failing with 2 == 4 before timing out [4].

5. Root-Cause Taxonomy and Factorial Program (Antigravity & Codex)

Independent post-hoc audits by Antigravity and Codex [8] formalize why the 7.1% resolution rate and 90.5% EBT non-utilization reflect an upstream floor effect:

- **H1 (Reasoning Token Starvation):** Unbounded thinking in shared 8,192-token budgets displaces JSON actions (11/67 calls = 16.4% of calls, 87.3% of output tokens). Prediction: Capping reasoning at 2,048 tokens or raising num_predict to ≥ 16,384 cuts truncation below 10%; Codex Study A tests this via a 2×2 (v4/v5 × think=false/true) design [8].
- **H2 (Pre-Training Prior Dominance):** Under cognitive load, 12B models default to pre-trained pytest assertions when unrestricted file writing is allowed (19/21 Cubie runs without EBT), particularly on stateful tasks vs. pure functions (FastAPI in 180702). Prediction: Dedicated contract tools raise EBT execution above 75% [8].
- **H3 (Baseline Pollution & Edit Friction):** The 65 pre-existing failures in Rich act as a noise attractor (0/14 solved). Prediction: Scoping run_tests to target paths (Codex Study D) and replacing text anchors with opaque file handles (Codex Study B) halves stalling and edit conflicts [8].
- **H4 (Model Capability Threshold & Human-in-the-Loop Value):** Evaluating a 30B+ model under identical freezes (Codex Study C) is predicted to lift resolution above 30% and meet the campaign qualification gate (≥ 2 independent task groups resolved per arm with red–green verification) [4, 8]. In human-in-the-loop usage on internal projects, the Project Lead observes that EBT contracts substantially improve test readability and oracle review speed [4].

6. Multi-Agent Follow-Up Study (Generations v6–v9)

To test whether role specialization overcomes single-agent overload, a follow-up study on November 12 evaluated four serial multi-agent generations (v6–v9: Analysis, QA, and Dev sharing the workspace under gemma4:12b, Seed 42, think=false, temperature=0.15, num_ctx=32768, num_predict=8192, and up to 1,800 s budget) [10]:

Generation & Archive	Exec. / Plan	Model / Tool Calls	Tokens (In / Out)	Reached Roles & Termination Cause
v6 (20261112-002038)	6 / 6	66 / 65	422,512 / 16,749	Analysis (100%) · finished: 6× stagnation stop (0/6 resolved)
v7-1 (20261112-013246)	3 / 6	46 / 45	316,839 / 5,649	Analysis (18), QA (27) · failed: post-freeze ValueError (0/3 resolved)
v7-2 (20261112-015457)	1 / 6	81 / 80	638,637 / 5,705	Analysis (100%) · interrupted: find_paths loop (0/1 resolved)
v8 (20261112-030527)	1 / 6	64 / 63	195,452 / 2,302	Analysis (61), QA (2) · interrupted: 30+ cursor_error loop (0/1 resolved)
v9 (20261112-035522)	6 / 6	1,184 / 1,179	6,980,998 / 44,845	Analysis (100%) · finished: 6× budget_exhausted, 974 dup. (0/6 resolved)
Total (v6–v9)	17 / 30	1,441 / 1,432	8,554,438 / 75,250	Analysis dominant · 0 / 17 privately resolved (0.0%)

6.1 Error Domain Migration Across the Four Multi-Agent Generations (v6–v9)

Across all 1,432 tool actions in v6–v9, 90.7% were search_content (795, 55.5%) and read_file (504, 35.2%), alongside 72 read_evidence (5.0%), 52 find_paths (3.6%), 2 ask_analysis (0.1%), 3 finish_orientation (0.2%), 3 create_case (0.2%), 1 collect_tests (0.1%), and 0 finish calls (0.0%) [10]:

1. **v6 (Archive 20261112-002038) — Two-Step Handoff Barrier:** Setting think=false restored JSON stability (65/66 valid calls; one 8,192-token regex loop in Rich repaired). Yet all 6 runs ended via stagnation stops (45–531 s) in orient without reaching QA or Dev; searching requests/utils.py instead of src/requests/utils.py returned empty hits, and the model never initiated publish_context followed by handoff, confirming Counter-Hypothesis V6-H8 [10].
2. **v7 (Archives 20261112-013246 & 20261112-015457) — Context Pollution & Tool-Removal Loops:** v7-1 reached QA (27 calls, 3 cases, 1 collection) before aborting on a post-freeze edit check. In v7-2, raw JSON ledgers and Cubie tutorials in the Analysis prompt truncated 54–55 tool turns; after recovery removed search_content, the agent invoked find_paths in repetitive loops (32 consecutive calls with "fastapi/routing.py" in the raw archive log, alongside 19 repeated "def get_route_info" pattern queries in the step-0073 request history) [10].
3. **v8 (Archive 20261112-030527) — Forced Handoff vs. Amnesia & Cursor Traps:** A deterministic controller rule (repeat_count >= 3 activating QA) triggered QA, which sent two ask_analysis queries (job-12, job-20). Analysis found the exact target in fastapi/dependencies/utils.py (lines 253–280), but because it did not call update_context, the read files were lost (materials: [] in job-20), while passing "cursor": "0" triggered 30+ consecutive cursor_error rejections [10].
4. **v9 (Archive 20261112-035522) — Infrastructural Maturity vs. Controller-Prompt Lock-In:** v9 introduced transactional Zero-Admin Memory, cursor-free schemas, symmetric read/search rights, and clean Markdown prompts, while removing v8's forced handoff [10]. Over 1,184 model calls and 1,179 tool steps (6.98M input tokens), infrastructural errors dropped to 0 cursor errors, 0 tool errors, and 0 context overflows, and Analysis located the target files within steps 0–13 on every task. However, removing the forced handoff while describing finish primarily with review parameters (verdict=verified/rework, raising ValueError('Orientation cannot verify a repair') in orient at team_v9.py:381) caused the model to invoke finish zero times (0 calls) across 1,184 steps and fall into 974 duplicate actions (82.6%). Because team_v9.py:615 injects the framework guide only upon entering QA, the qb-cubie instructions were never presented to the model in any v9 run [10].

7. Conclusion

Across nine agent generations (59 executed runs: 42 single-agent in v1–v5 and 17 multi-agent in v6–v9), we contribute an open-source declarative testing framework (qb-cubie 0.0.3), a zero-leakage SWE-bench evaluation harness, and forensic evidence of the floor effects governing 12B models on everyday hardware. While flat single-agent architectures achieved three verified repairs—including a proof-of-concept resolution executing six EBT contracts in archive 180702—upstream bottlenecks (87.3% reasoning token starvation in v5, 90.5% EBT non-utilization from pytest priors, and phase-transition lock-in in v6–v9) prevented downstream test methodologies from being systematically exercised. Overcoming this floor requires decoupling reasoning budgets, enforcing dedicated contract-authoring tools, and pairing unambiguous phase transitions with higher-capacity models.

8. References and Frozen Study Artifacts

1. **Hoare Logic:** Hoare, C. A. R. (1969). An axiomatic basis for computer programming (`{P} S {Q}` pre-state, operation, and post-state triple referenced in Section 1.1). *Communications of the ACM*, 12(10), 576–580.
2. **SWE-bench Suite:** Jimenez, C. E., Yang, J., Wettig, A., Yao, S., Pei, K., Press, O., & Narasimhan, K. (2024). SWE-bench: Can Language Models Resolve Real-World GitHub Issues? (SWE-bench, SWE-bench Lite, and SWE-bench Verified referenced in Section 2). *ICLR 2024*.
3. **Agentic Harnesses:** Yang, J., et al. (2024), SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering; Zhang, Y., et al. (2024), AutoCodeRover: Autonomous Program Improvement; Xia, C. S., et al. (2024), Agentless: Demystifying LLM-based Software Engineering Agents (referenced in Section 2).
4. **Study Protocol & Core Reports (2026):** Wirzba, D., *qb-cubie in a Test-Guided Coding Agent* (`PAPER.md`, Nov 11, 2026; v1–v5), *Study Protocol* (`PROTOKOLL.md`), and *Container Replay Validation* (`VALIDIERUNG.md`) in `experiments/testguided-study/`.
5. **Task Pool & Candidate Audits (2026):** *SWE-bench Task Pool Documentation* (`DATEN.md`), *Individual Task Audits* (`KANDIDATEN.md`), and `candidate-catalog.json` in `experiments/testguided-study/`.
6. **Prompts & Single-Agent Designs (2026):** Agent guides (`guides/cubie.md`, `pytest.md`) and *Single-Agent Designs v1–v5* (`AGENTENDESIGN.md`, `V4.md`, `V5.md`) in `experiments/testguided-study/`.
7. **Single-Agent Pilot Analyses (2026):** *Pilot Analyses v2–v4* (`PILOTANALYSE-20261111.md`, `192412.md`) and *Visible Baseline Technical Analysis* (`TECHNIKANALYSE-20261111-171515.md`) in `experiments/testguided-study/`.
8. **Root-Cause Audits (2026):** Antigravity, *Cascade Hypothesis System H1–H4* (`HYPOTHESEN-URSACHENANALYSE-20261111.md`, Gemini-3.8-Flash) and Codex, *Workflow Bottleneck & Factorial Program* (`URSACHENANALYSE-CODEX-20261111.md`, GPT-6.1-Sol).
9. **qb-cubie 0.0.3 Guides & Technical Audit (2026):** Documentation in `docs/GUIDE_*.md` (*Contract Semantics*, *Quality Assurance*, *EBT*, *RBT*, *Arena Management*, *Linters*, *Benchmarks*), `src/qb_cubie/benchmarks/swebench/`, and *Critical Technical Audit* (`./GUTACHTEN.md`, commit `5e50d22f80db90c4e8ae6e3aeedab6acdb981230`).
10. **Multi-Agent v6–v9 & Submission Audits (2026):** *Multi-Agent Designs & Implementations v6–v9* (`AGENTENDESIGN-V6.md` to `V9.md`, `IMPLEMENTIERUNG-V7.md`, `V9.md`), *v6 & v9 Pilot Analyses*, *Comparative Synthesis v6–v9* (`ANALYSE-GENERATIONEN-V6-V9.md`), and *Submission Audits* (`KRITISCHE-ANALYSE-PAPER-SUBMISSION.md`, Stage 1 & 2 Grounding Audits, Nov 12, 2026).